

Microgrid Operations and Incident Response Guide

A practical reference for operators of the Northstar Grid solar, battery, and backup-generator system.

Purpose

This guide explains normal operating targets, escalation rules, and first-response actions for a site microgrid. It is designed for control-room operators and on-call technicians. It does not replace electrical safety procedures or local emergency requirements.

System at a glance

Asset	Normal role	Operating boundary
Solar array	Primary daytime generation	Curtail only when instructed by the energy controller
Battery energy storage system (BESS)	Smooths demand and provides short outage support	Normal state of charge: 35% to 90%
Backup generator	Extended outage and low-state-of-charge protection	Automatic start threshold: 20% state of charge
Energy management system (EMS)	Schedules assets and monitors alarms	Operator must acknowledge critical alarms within 10 minutes

Operating principle: protect people first, preserve critical loads second, and optimize energy cost only after the system is stable.

Key terms

State of charge (SoC): battery energy available as a percentage. Island mode: the site is operating without utility-grid support. Critical loads: safety, communications, refrigeration, and designated server-room circuits.

1. Normal operation and operating modes

The EMS normally selects the operating mode automatically. Operators should confirm the selected mode during every shift handover and whenever an alarm is raised.

Mode	When used	Operator action
Grid-connected	Utility service is available	Monitor SoC, solar output, and import demand. No manual action is normally required.
Peak-shaving	Site demand approaches the contracted import limit	Verify battery discharge is within schedule. Escalate if import limit is exceeded for 15 minutes.
Island mode	Utility outage or planned isolation	Confirm critical-load priority and watch SoC every 15 minutes.
Generator support	SoC reaches 20% or outage is expected to exceed 2 hours	Confirm generator start, fuel status, and stable frequency before reducing monitoring frequency.

Battery thresholds

SoC range	Meaning	Required response
90% to 100%	High charge	EMS may curtail solar. Check that curtailment is expected.
35% to 90%	Normal band	Continue normal monitoring.
20% to 34%	Low reserve	Notify on-call technician and verify generator readiness.
Below 20%	Critical reserve	Generator should start automatically. Escalate immediately if it does not.

Shift handover checklist

1. Confirm the active mode. 2. Record SoC and grid status. 3. Review unresolved alarms. 4. Confirm generator fuel is above 50%. 5. Confirm the next planned maintenance window.

Important: Do not manually force battery charging or discharging during a utility outage unless directed by the on-call electrical engineer.

2. Alarm response and incident escalation

An alarm is not automatically an incident. An incident begins when safety, critical loads, equipment integrity, or service continuity is at risk.

Alarm severity and first response

Severity	Examples	First response	Escalation
Warning	Reduced solar output; communication delay	Acknowledge and investigate during the shift.	Escalate if unresolved after 4 hours.
Major	SoC below 35%; inverter unavailable	Check operating mode and critical-load status.	Notify on-call technician within 30 minutes.
Critical	Generator fails to start; BESS thermal alarm; critical load lost	Protect people, preserve critical loads, and record time of event.	Call on-call engineer immediately and notify site manager.

Utility outage procedure

Step 1: Confirm utility loss in the EMS and site switchboard indicator. Step 2: Confirm island mode within five minutes. Step 3: Check SoC and critical-load status. Step 4: If SoC is below 35%, alert the on-call technician. Step 5: If SoC reaches 20%, verify generator start. Step 6: Record the incident timeline until utility service is restored.

Battery thermal alarm

Do not open battery cabinets. Keep personnel away from the BESS enclosure, follow the site emergency plan, and contact the electrical engineer immediately. If smoke, fire, or a strong chemical odor is present, activate site emergency procedures and call emergency services.

Incident record

Record	Minimum detail
Timeline	Alarm time, acknowledgement time, mode changes, calls made, restoration time
Evidence	Alarm code, SoC, screenshots, generator runtime, affected loads
Outcome	Cause if known, temporary action, owner, follow-up due date

3. Maintenance, data quality, and FAQs

Reliable RAG testing benefits from precise operating facts, exceptions, and intentionally unanswered questions. This page provides additional reference material for that purpose.

Preventive maintenance schedule

Frequency	Task	Owner
Each shift	Review alarms, SoC, grid mode, and generator fuel level	Operator
Monthly	Inspect BESS enclosure externally and review inverter event logs	Technician
Quarterly	Run generator under load for 30 minutes and verify automatic transfer	Electrical engineer
Annually	Verify protection settings and complete thermal inspection	Qualified contractor

Data-quality checks

If solar production reads zero during daylight, first check weather conditions, inverter status, and telemetry freshness. Treat telemetry older than 15 minutes as stale. Do not make control decisions using stale telemetry without confirming local indicators or contacting the technician.

Frequently asked questions

Question	Answer
When should the generator start?	Automatically at 20% SoC, or earlier when an outage is expected to exceed two hours.
Who acknowledges critical alarms?	The control-room operator must acknowledge them within 10 minutes and call the on-call engineer immediately.
Can an operator open a BESS cabinet during an alarm?	No. Operators must not open battery cabinets during normal operations or a thermal alarm.
What is the cryptocurrency reimbursement policy?	This guide does not contain a cryptocurrency reimbursement policy.

Document boundary: This guide does not specify generator fuel-purchase vendors, employee travel reimbursement, or cryptocurrency payments. A grounded assistant should state that these topics are not covered rather than inventing a policy.